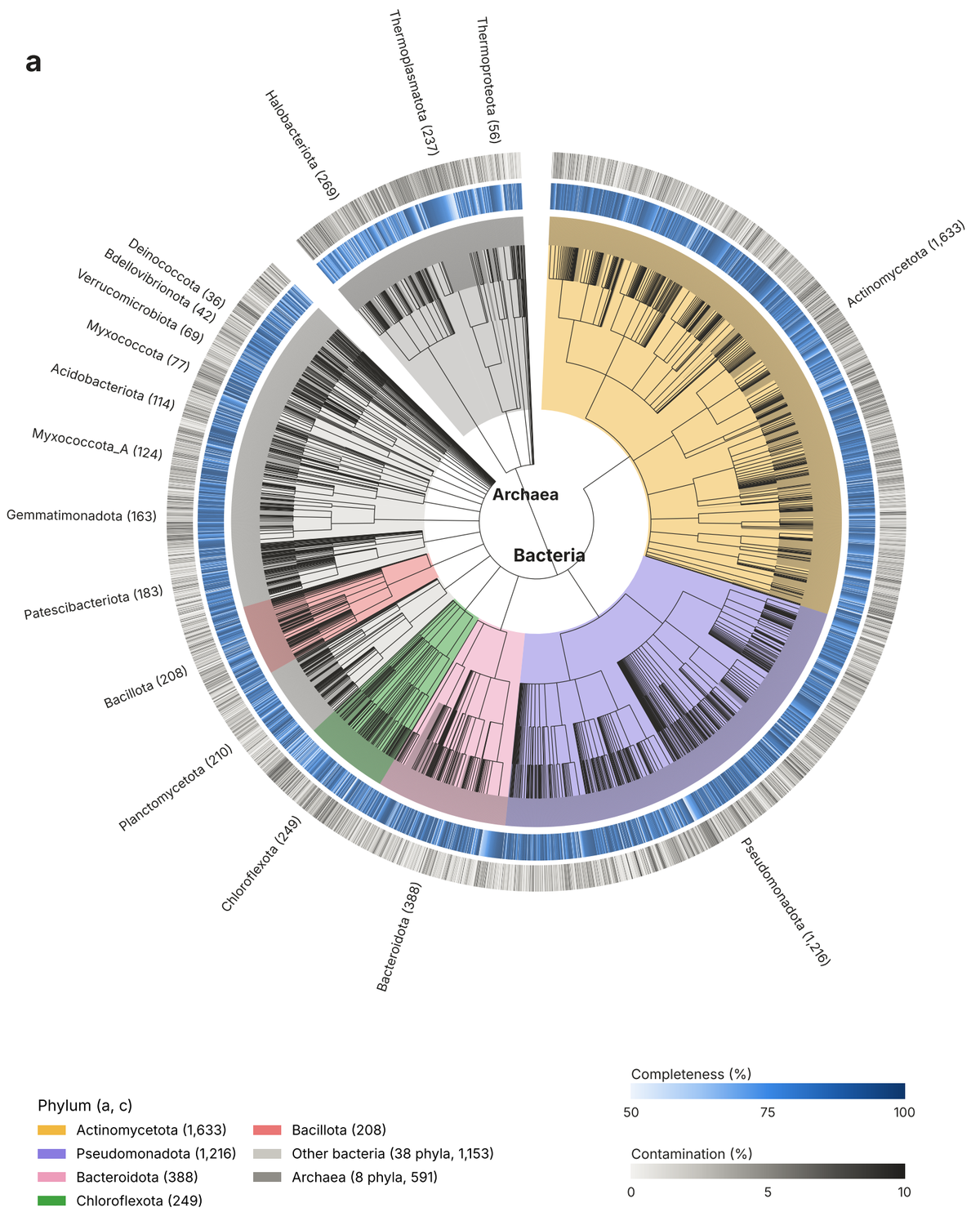
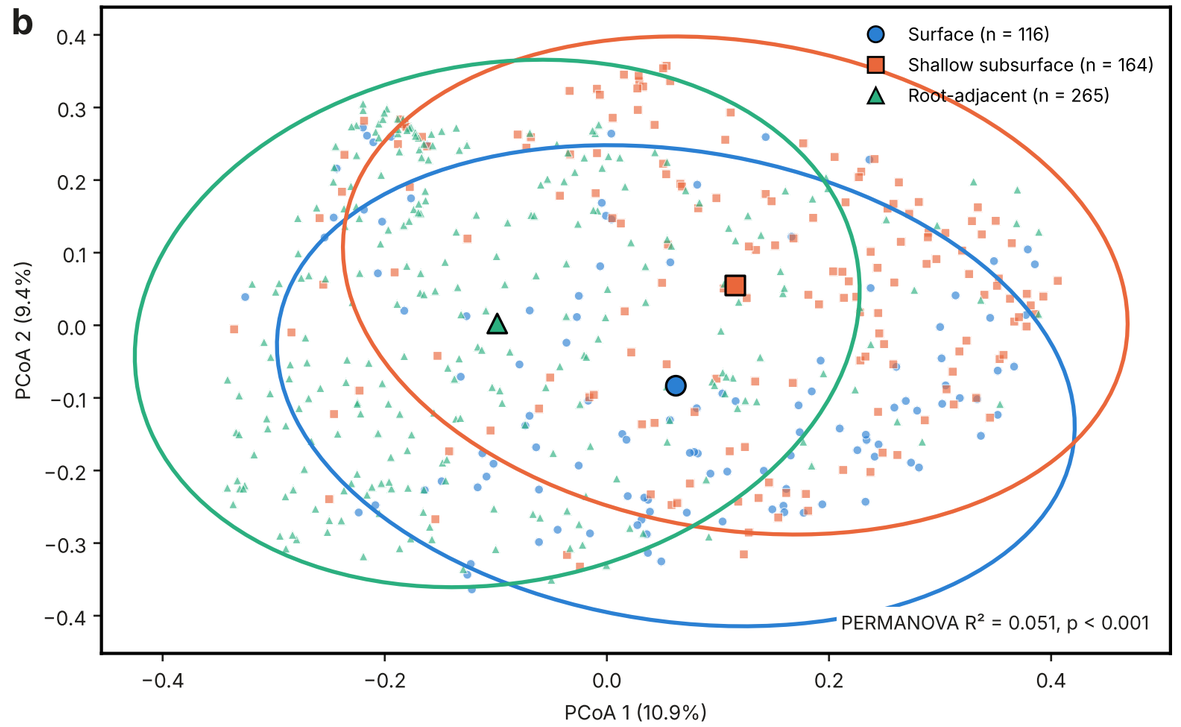
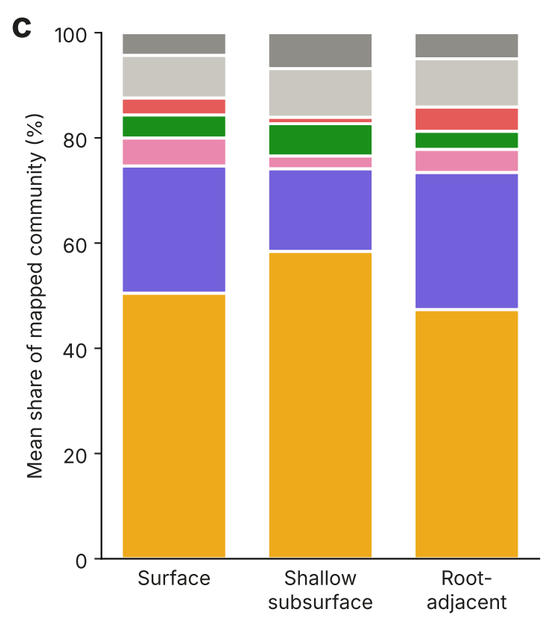
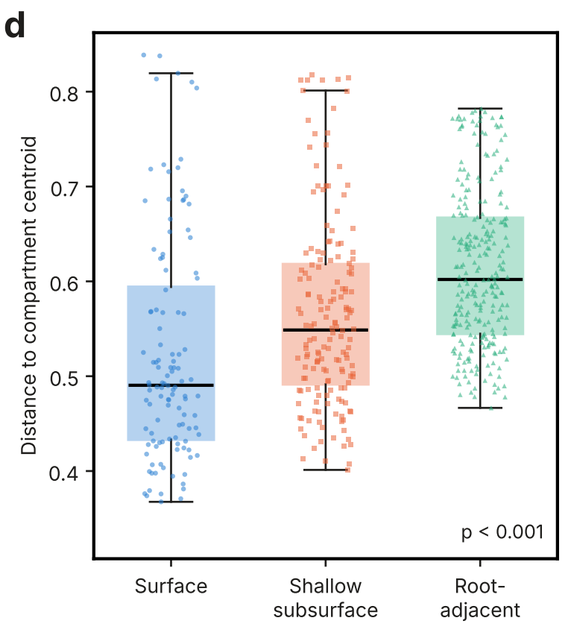

a
Halobacteriota (269)
Thermoplasmatota (237)
Thermoproteota (56)
Actinomycetota (1,633)
Deinococcota (36)
Bdellovibrionota (42)
Verrucomicrobiota (69)
Myxococcota (77)
Acidobacteriota (114)
Myxococcota_A (124)
Gemmatimonadota (163)
Patescibacteriota (183)
Bacillota (208)
Planctomycetota (210)
Chloroflexota (249)
Bacteroidota (388)
Pseudomonadota (1,216)
Archaea
Bacteria
Completeness (%)
50
75
100
Contamination (%)
0
5
10
Phylum (a, c)
Actinomycetota (1,633)
Pseudomonadota (1,216)
Bacteroidota (388)
Chloroflexota (249)
Bacillota (208)
Other bacteria (38 phyla, 1,153)
Archaea (8 phyla, 591)


b
Surface (n = 116)
Shallow subsurface (n = 164)
Root-adjacent (n = 265)
PERMANOVA R² = 0.051, p < 0.001
PCoA 2 (9.4%)
PCoA 1 (10.9%)
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.4
−0.2
0.0
0.2
0.4


c
Mean share of mapped community (%)
100
80
60
40
20
0
Surface
Shallow subsurface
Root-adjacent


d
Distance to compartment centroid
0.8
0.7
0.6
0.5
0.4
p < 0.001
Surface
Shallow subsurface
Root-adjacent